

INDIA SPACE LAB

BA/14B, Janakpuri, New Delhi, India

COMPREHENSIVE PROJECT REPORT

- Project 1: CanSat Ground Control Software (GCS)
- Project 2: Disaster Management — Heat Wave Study
- Project 3: Rocketry — FEM + CFD Simulation (SimScale)
- Project 4: Advanced Drone Technology — PID & Guidance

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PROJECT 1: CanSat Ground Control Software (GCS)

1.1 Project Overview

This project requires students to design and develop a professional single-page Ground Control Software (GCS) for a CanSat mission. The software monitors telemetry data in real time, visualizes mission parameters, displays GPS tracking, handles mission controls, and simulates aerospace mission operations. The interface must be modular, readable, and mission-oriented, similar to professional aerospace systems.

1.2 Interface Design & Components

1. Interface Layout

Single-page dashboard divided into: Telemetry, Graphs, Controls, Tracking Map, Orientation Visualization, and Video Stream. Tools: HTML5, CSS3, CSS Grid, Flexbox, JavaScript.

2. Top Control Bar

Mission operation controls: Start/Stop telemetry, Export CSV, Export Graph, Sync PC Time, Reset Packet. Tools: JavaScript DOM Manipulation.

3. Mission Control Panel

Critical operations: Manual Separation, Emergency Parachute Deployment, Redundant Activation. Dynamic command execution status display. Tools: JavaScript, Web Serial API.

4. Telemetry Display

Continuous parsing and display of telemetry packets — container and payload separately. Tools: WebSocket, Web Serial API, JavaScript String Parsing.

5. Real-Time Graphs

Continuously updating graphs for Altitude, Pressure, Temperature, Descent Rate, and Battery Voltage using Chart.js or similar libraries.

6. Tracking Map

Live GPS coordinates displayed on Leaflet.js + OpenStreetMap map with trajectory history.

7. Orientation Visualization

Dynamic Roll/Pitch/Yaw visualization using Three.js — artificial horizon or 3D model.

8. Live Video Streaming

Browser camera API integration with camera selection, start/stop controls, and status indication.

9. Data Management

Telemetry logging, CSV export, graph export, packet reset, and organized telemetry storage.

1.3 Error Code System

A 4-digit error code system identifies mission faults. Each digit represents a specific condition: 0 = normal, 1 = fault. The digits monitor descent rate, GPS availability, payload separation, and emergency parachute status.

Digit	Condition	0 = Normal	1 = Fault
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1	Descent Rate	Within 8–10 m/s	Outside safe range
2	GPS Availability	GPS data available	GPS data unavailable
3	Payload Separation	Separated successfully	Separation failure
4	Emergency Parachute	Parachute inactive	Parachute activated

Examples: 0000 = All normal | 1000 = Descent fault | 0100 = GPS unavailable | 0010 = Payload failure | 0001 = Parachute activated | 1111 = All faults active

1.4 Deliverables & Evaluation Criteria

- Ground Control Software Source Code
- Executable/Application Files
- Project Documentation/Report
- UI Screenshots
- Demonstration Video
- Sample Telemetry Logs
- Exported CSV Samples
- Graph Export Samples

Evaluation Criteria	Weightage
UI/UX Design	15%
Telemetry Handling	20%
Real-Time Visualization	20%
Mission Control Features	15%
Graphing and Tracking	10%
Orientation and Video Systems	10%
Code Quality and Scalability	10%

PROJECT 2: Disaster Management — Heat Wave Study

2.1 Introduction & Objectives

Title: "Heat Wave Response, Recovery, Future Challenges, and the Role of Technology: A Ground-Level Assessment and Preparedness Study"

Heat waves are emerging as one of the most dangerous climate-related disasters in India. Increasing temperatures due to climate change, rapid urbanization, deforestation, and changing land-use patterns have significantly increased the frequency and severity of extreme heat events. This project targets PG students, interns, disaster management professionals, and researchers.

Primary Objectives:

- Understand the causes and impacts of heat waves
- Analyze response mechanisms adopted during heat waves
- Study recovery measures after heat-related disasters
- Identify future challenges associated with rising temperatures
- Study urban heat island effects and evaluate community awareness
- Suggest technological and policy improvements

2.2 Methodology & Data Collection

Data collection combines primary surveys (field visits, interviews, questionnaires, temperature observations, focus group discussions) with secondary sources (IMD records, NDMA guidelines, WHO reports, municipal and health department data).

Suggested Vulnerable Groups:

Traffic Police	Street Vendors
Farmers	Construction Workers
Elderly People	Children
Slum Residents	Sanitation Workers

2.3 Key Components & Role of Technology

Early Warning Systems: IMD forecasting, SMS alerts, mobile apps, AI-based prediction

GIS and Remote Sensing: Thermal hotspot mapping, satellite temperature monitoring, urban heat mapping

Artificial Intelligence: Heat prediction models, risk assessment, health risk forecasting

Internet of Things (IoT): Smart temperature sensors, smart cooling systems, automated weather stations

Drones: Monitoring vulnerable areas, survey during emergencies

Smart Infrastructure: Cool roofs, reflective pavements, green buildings, smart irrigation

2.4 Deliverables & Report Structure

Deliverables:

- Project Report
- Survey Data Sheets
- GIS/Temperature Maps

- Charts and Graphs
- Case Studies
- Photographs
- Recommendations
- PowerPoint Presentation

Chapter	Title
Ch. 1	Introduction to Heat Waves
Ch. 2	Literature Review
Ch. 3	Study Area Profile
Ch. 4	Research Methodology
Ch. 5	Heat Wave Response Analysis
Ch. 6	Recovery Measures
Ch. 7	Future Challenges
Ch. 8	Role of Technology
Ch. 9	Findings and Recommendations
Ch. 10	Conclusion

Activity	Duration
Literature Review	1 Week
Field Survey	2 Weeks
Data Collection	1 Week
Data Analysis	1 Week
Report Writing	1 Week
Presentation	1 Day

PROJECT 3: Rocketry — Aerospace Simulation (FEM + CFD)

3.1 Theoretical Background (Part A)

Q1 — Finite Element Method (FEM)

Explain FEM in your own words: what it is, why it is used, and its applications in aerospace engineering.

Q2 — Von Mises Stress

Define Von Mises stress and explain its importance for yielding and structural failure analysis.

Q3 — Role of Meshing

Discuss how mesh density affects accuracy, computational cost, and simulation mesh dependency.

Q4 — CFD vs FEM

Differentiate between Computational Fluid Dynamics and Finite Element Method: compare objectives, outputs, and engineering applications.

Q5 — Boundary Conditions

Explain the importance of boundary conditions (supports, loads, constraints) and the impact of incorrect assumptions.

3.2 FEM Project Tasks (Part B)

Task 1 — Rocket Fin Geometry & Material

Create or import rocket fin geometry in SimScale. Assign appropriate material properties (e.g., aluminium alloy or carbon fibre).

Task 2 — Boundary Conditions

Apply realistic fixed supports at fin root and aerodynamic pressure loads on surfaces.

Task 3 — Mesh Study

Perform coarse, medium, and fine mesh analysis. Compare stress and deformation results across mesh densities.

Task 4 — Result Interpretation

Analyze Von Mises stress distribution, total deformation, and identify critical failure regions.

Task 5 — Factor of Safety

Calculate Factor of Safety ($\text{FoS} = \text{Yield Strength} / \text{Max Von Mises Stress}$). Determine structural safety.

3.3 CFD Project Tasks (Part C)

Task 1 — Flow Domain Creation

Create the external flow region (bounding box) around the rocket/fin geometry in SimScale.

Task 2 — CFD Boundary Conditions

Apply inlet velocity (representing flight speed), outlet pressure, and wall no-slip conditions.

Task 3 — CFD Mesh Generation

Generate the computational mesh and explain the refinement strategy (surface refinement, boundary layers).

Task 4 — Flow Visualization

Analyze and visualize pressure contours, velocity contours, and streamlines around the geometry.

Task 5 — Aerodynamic Discussion

Discuss drag forces, wake formation behind the fin, and implications for aerodynamic stability.

BONUS: Modify the fin design to reduce stress or aerodynamic drag. Compare original vs optimized design using proper engineering analysis and SimScale screenshots.

3.4 Report Structure & Evaluation Criteria

Recommended Report Sections: Aim | Theory | Geometry | Material Properties | Boundary Conditions | Mesh Details | Results | Discussion | Conclusion | References

Evaluation Criteria	Weightage
Understanding of Concepts	20%
Simulation Setup	20%
Result Interpretation	25%
Engineering Reasoning	25%
Report Presentation	10%

Note: Simulation results are highly dependent on assumptions, mesh quality, material properties, and boundary conditions. Engineering judgment must always accompany simulation results.

PROJECT 4: Advanced Drone Technology — PID Control & Guidance

4.1 Objectives & Environment Setup

- Understand the fundamentals of Guidance and Control for autonomous moving systems
- Learn PID (Proportional–Integral–Derivative) controller tuning basics
- Observe the effect of environmental disturbances (wind/current) on stability
- Develop intuition behind feedback-based control systems
- Understand trajectory/path-following concepts used in drones and boats
- Gain hands-on experience with Python-based simulations using Google Colab

Environment: Google Colab (Python-based, no local installation required). Notebook link: colab.research.google.com. Alternative: run locally in VS Code, Spyder, Jupyter Notebook, or Anaconda.

4.2 Task 1 — PID Controller (Drone Altitude Control)

The drone attempts to reach and maintain a desired altitude. Students tune PID gains and observe system response with and without wind disturbance.

- Run the code cell and use sliders to adjust P, D, I gains and Wind Disturbance
- No Wind Zone: 0–6 seconds | Wind Disturbance introduced after 6 seconds
- Tuning order: First tune P gain → then D gain → finally I gain
- Target performance: Faster response, Minimum overshoot, Reduced oscillation, Better disturbance rejection
- Run animation cell to view drone motion: observe stability, oscillations, recovery after disturbance

Gain	Effect When Increased	Effect When Decreased
Proportional (Kp)	Faster response, more overshoot	Slower response, less overshoot
Integral (Ki)	Eliminates steady-state error	Steady-state error remains
Derivative (Kd)	Reduces oscillations/overshoot	More oscillations

4.3 Task 2 — Guidance & Path Tracking (Autonomous Boat)

The autonomous boat follows a predefined trajectory. Students tune controller gains and environmental current values to achieve accurate path tracking.

- Run notebook cells sequentially
- Use sliders to tune: kp, kd, current_x, current_y
- Blue dotted line = desired trajectory | Boat trajectory = actual tracking
- Observe: tracking accuracy, oscillations, path deviation, smoothness of motion
- Update final cell parameters with tuned values and run animation
- Without disturbance case: set current_x = 0, current_y = 0

BONUS — Figure-Eight Drone Simulation: Create a Python simulation where a drone follows a figure-eight trajectory using waypoint tracking or parametric equations. Bonus marks for smooth trajectory, stable control, good animation, and creativity.

4.4 Deliverables

- Completed .ipynb or .py file with all simulation code

- PID Controller Tuning Result Plot (drone altitude control)
- Boat Guidance Plot — with disturbance/current
- Boat Guidance Plot — without disturbance/current (current_x=0, current_y=0)
- (Optional) Figure-Eight Drone Simulation animation

5. CONCLUSION & SUMMARY

This India Space Lab Project Report (May–June 2026) encompasses four comprehensive hands-on projects spanning aerospace engineering, disaster management, structural simulation, and autonomous systems control.

Project	Domain	Key Skills	Tools
CanSat GCS	Avionics / Software	Web dev, Telemetry, Real-time systems	HTML5, JS, Chart.js, Leaflet.js, Three.js
Heat Wave Study	Disaster Management	Field research, GIS, Data analysis	GPS, GIS, IMD data, Survey tools
Rocketry Simulation	Aerospace / Structural	FEM, CFD, Engineering analysis	SimScale, CAD software
Drone & Guidance	Control Systems	PID tuning, Path tracking, Python	Python, Google Colab, Matplotlib

Each project is designed to build practical competency at the intersection of space technology, data science, engineering simulation, and autonomous systems — preparing students for careers in the growing Indian and global space ecosystem.